

Communication System:

Assign - 4.

①. $G = 20 \text{ dB}$ $B = 10^6 \text{ Hz}$

i/p signal power = $2 \times 10^{-6} \text{ W}$

$N_{\text{PSD}} = 2 \times 10^{-15} \text{ W/Hz}$.

a). $N_i = N_{\text{PSD}} \times B = 2 \times 10^{-15} \times 10^6 = 2 \times 10^{-9} \text{ W}$.

b). $G_{\text{dB}} \Rightarrow 20 = 10 \log_{10} g$.

$S_o = 2 \times 10^{-6} \times 100$ $g = 100$
 $= 200 \text{ mW}$.

c). $N_o = N_i \times g = 200 \text{ mW}$

d). $\text{SNR}_{i/p} = \frac{S_i}{N_i} = \frac{2 \times 10^{-6}}{2 \times 10^{-9}} = 1000$

$\text{SNR}_{o/p} = \frac{S_o}{N_o} = \frac{200 \times 10^{-6}}{200 \times 10^{-9}} = 1000 (30 \text{ dB})$

e). $F = \frac{\text{SNR}_i}{\text{SNR}_{op}} = 1$

2) $S_i = 5 \text{ MW}$ $N_i = 0.05 \text{ MW}$, $S_o = 200 \text{ MW}$
 $N_o = 8 \text{ MW}$

a) $\text{SNR}_i = \frac{S_i}{N_i} = \frac{5 \times 10^{-6}}{0.05 \times 10^{-6}} = 100$

$\text{SNR}_o = \frac{S_o}{N_o} = \frac{200 \times 10^{-6}}{8 \times 10^{-6}} = 25$

$F = \frac{\text{SNR}_i}{\text{SNR}_o} = \frac{100}{25} = 4.$

b) $\text{NF(dB)} = 10 \log_{10} F = 10 \log_{10} 4 = 6.02 \text{ dB}$

3) $G_1 = 10^{12/10} \approx 15.85$ $F_1 = 10^{3/10} \approx 1.995$

$G_2 = 10^{10/10} = 10$, $F_2 = 10^{6/10} = 3.981$

$F_3 = 10^{7/10} = 10^{0.7} \approx 5.012$

a) $F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$
 $\approx 1.995 + \frac{3.981 - 1}{15.85} + \frac{5.012 - 1}{(15.85)(10)} \approx 2.208$

b) $\text{NF}_{\text{total}} = 10 \log 2.208 = 3.44 \text{ dB}$

4) a) True, $\text{NF(dB)} = \text{SNR}_{in}(\text{dB}) - \text{SNR}_{out}(\text{dB})$
 $\text{SNR}_{out} = \text{SNR}_{in} - \text{NF}$ $\text{NF} = \text{dB}$
 $= \text{SNR}_{in} - 6$

b) True. $\text{SNR} = \frac{S}{N_{in}} = \frac{S}{N_{out} B}$ $B' = 2B$

$\text{SNR}' = \frac{S}{N_{out}(2B)} = \frac{1}{2} \left(\frac{S}{N_{out} B} \right) = \frac{1}{2} \text{SNR}.$

5) $\text{Smf}(t) = S(T-t)$

$S(t) = \mathcal{I}_{[0,2]}(t) - 2 \mathcal{I}_{[1,3]}(t)$, signal end $T=3$

$S(t) = \begin{cases} 1 & 0 \leq t < 1 \\ -1 & 1 < t < 2 \\ -2 & 2 < t \leq 3 \\ 0 & \text{otherwise} \end{cases}$

$\text{Smf}(t) =$

$\text{Smf}(t) = S(3-t).$

$\text{Smf}(t) = \begin{cases} 1 & 2 < t \leq 3 \\ -1 & 1 \leq t \leq 2 \\ -2 & 0 \leq t < 1 \end{cases}$

$\text{Smf}(t) = -2 \mathcal{I}_{[0,1]}(t) - \mathcal{I}_{[1,2]}(t) + \mathcal{I}_{[2,3]}(t).$

b) $Z(t) = s(t) * \text{smf}(t)$
 $t = T = 3, Z(3) = \int_0^3 |s(t)|^2 dt = \int_0^1 1^2 dt + \int_1^2 (-1)^2 dt + \int_2^3 -2^2 dt$
 $1 + 1 + 4 = 6.$

c) $h(t) = \mathcal{I}_{[0,1]}(t)$ $y(t) = (x * h)(t)$ $z(t) = (x * \text{smf})(t)$
 $\text{smf}(f) = -2 \cdot h(f) - 1 \cdot h(f-1) + 1 \cdot h(f-2)$
 $z(t) = x(t) * [-2h(t) - h(t-1) + h(t-2)]$
 $= -2y(t) - y(t-1) + y(t-2).$

6) $d = 36,000 \text{ km} = 36 \times 10^7 \text{ m}.$

$P_t = 100 \text{ W} / 20 \text{ dBW}.$ $G_t = 18 \text{ dB}$

$\text{EIRP} = 38 \text{ dBW} = P_t(\text{dBW}) + G_t(\text{dB})$ $f = 4 \text{ GHz}$

$D = 3 \text{ m}$ $\eta = 0.5$

$d = \frac{c}{f} = 0.075 \text{ m}$ $A = \frac{\pi D^2}{4} \approx 7.0686 \text{ m}^2$

$G_R = \frac{4A^2}{\lambda^2} = \frac{0.5 \times 7.0686 \times 4\pi}{(0.075)^2} \approx 78956$

$G_R(\text{dB}) \approx 38.97 \text{ dB}$ $\text{FSPL} = \left(\frac{4\pi d}{\lambda}\right)^2 \approx 3.638 \times 10^9$

$\text{FSPL}_{\text{dB}} = 195.58 \text{ dB}.$

$P_R = \text{EIRP}_{\text{dBW}} + G_R(\text{dB}) - \text{FSPL}_{\text{dB}}$

$\Rightarrow 38 + 38.97 - 195.58 \approx 118.61 \text{ dBW}.$

$P_R = 10^{\frac{-118.61}{10}} \approx 1.38 \times 10^{-12} \text{ W} = 1.38 \text{ pW}.$

7) $X(f) = \int_0^1 1 - \frac{1f}{10} \quad |f| \leq 10 \text{ kHz}$
 otherwise 0

$f_s = 15 \text{ kHz}.$

1) $f_m = 10 \text{ kHz}$, $f_N = 2 \times f_m = 20 \text{ kHz}$
 (Nyquist rate). $f_s = 15 \text{ kHz}.$

$f_s < 2f_m \rightarrow$ not satisfied.

2) Baseband range $\Rightarrow -10 \rightarrow +10 \text{ kHz}$

$f_s = 15 \text{ kHz}$ 1st +ve range $[15-10, 15+10]$
 $= [5 \text{ kHz}, 25 \text{ kHz}]$

1st -ve range $[-15-10, -15+10] = [-25 \text{ kHz}, -5 \text{ kHz}]$

overlap range $5 \text{ kHz} < |f| < 10 \text{ kHz}.$

3) For perfect reconstruction $\rightarrow$ reconstructed must be identical to LPF with 10 kHz will pass

passband $x(t)$ signal but will include the aliased components in 5-10 kHz range.

$$8) 120 \leq |f| \leq 150 \text{ kHz} \quad f_L = 120 \text{ kHz} ; f_H = 150 \text{ kHz}$$

$$\textcircled{1} B = f_H - f_L = 30 \text{ kHz}$$

$$\textcircled{2} \frac{2f_H}{k} \leq f_s \leq \frac{2f_L}{k-1} \quad k=2$$

$$f_L > \frac{2 \times 150}{2} = 150 \text{ kHz}$$

$$f_H > \frac{2 \times 120}{2-1} = 240 \text{ kHz}$$

$$\textcircled{3} f_s = 80 \text{ kHz} \quad (k_{\min}) = \left\lceil \frac{f_H}{B} \right\rceil = \left\lceil \frac{150}{30} \right\rceil = 5$$

$$\text{for } k=4 \Rightarrow f_L = \frac{2(150)}{4} = 75 \text{ kHz}$$

$$f_H = \frac{2(120)}{3} = 80 \text{ kHz}$$

$f_s = 80 \text{ kHz}$ satisfy the condition for $k=4$.

$$9) f_s = 40 \text{ kHz} \quad 5 \leq |f| \leq 15 \text{ kHz}$$

$$f_{\text{alias}} = |f - k \cdot f_s|$$

$$f_{\text{alias-max}} = 15 \text{ kHz} \quad f_{\text{alias-min}} = 5 \text{ kHz}$$

$$B = 15 - 5 = 10 \text{ kHz}$$

$$2) f = k \cdot f_s \pm f_{\text{alias}}$$

$$k=1; f_L = 1(40) - 15 = 25 \text{ kHz} \quad f_H = 40 - 5 = 35 \text{ kHz}$$

$$f_L = 40 + 5 = 45 \text{ kHz}, f_H = 40 + 15 = 55 \text{ kHz} \quad (\text{inverted spectrum})$$

$$3) f_c = \frac{f_L + f_H}{2} = \frac{45 + 55}{2} = 50 \text{ kHz} \quad (\text{non-inverted spectrum})$$

$$10) x(t) = m(t) \cos(2\pi f_c t)$$

$$B = 8 \text{ kHz} ; f_c = 60 \text{ kHz}$$

$$f_L = f_c - B = 60 - 8 = 52 \text{ kHz} \quad f_H = f_c + B = 60 + 8 = 68 \text{ kHz}$$

$$f_s = 2 \times f_H = 2 \times 68 \text{ kHz} = 136 \text{ kHz}$$

$$3) f_s = \frac{2f_H}{k} \quad k \leq \frac{f_H}{B} \quad , B W = f_H - f_L = 68 - 52 = 16 \text{ kHz}$$

$$k = \frac{68}{16} = 4.25 \Rightarrow k=4$$

$$\text{min } f_s = \frac{2 \times 68}{4} = \frac{136}{4} = 34 \text{ KHz}$$

$$4) f_s = 25 \text{ KHz} \quad \text{falias} = |f - k f_s|$$

$$f_s = 25 \text{ KHz}$$

$$k=2 \Rightarrow 160 - 2(25) = 110 \text{ KHz}$$

$$[52, 68] \text{ shifted by } -50 \text{ KHz } (k=2)$$

$$\text{new range } (52-50, 68-50) = [2, 18] \text{ KHz}$$

$$f_c = 10 \text{ KHz}$$

11) N levels of range $[-V, V]$ $\sigma = \frac{2V}{N}$
for large # of levels N.

$$P_g = \frac{\Delta^2}{12} = \frac{(V/N)^2}{12} = \frac{4V^2}{12N^2} = \frac{V^2}{3N^2}$$

$$\alpha = \frac{V}{\sigma} \Rightarrow V = \alpha \sigma$$

$$P_g = \frac{(\alpha \sigma)^2}{3N^2} = \frac{\alpha^2 \sigma^2}{3N^2} \quad ; \quad \text{SNR} = \frac{\sigma^2}{P_g} = \frac{3N^2}{\alpha^2}$$

$$N = 2^R \Rightarrow \text{SNR} = \frac{3(2^R)^2}{\alpha^2} = \frac{3 \cdot 2^{2R}}{\alpha^2}$$

$$2) \text{SNR}_{dB} = 10 \log_{10} \left(\frac{3 \cdot 2^{2R}}{\alpha^2} \right)$$

$$\text{SNR}_{dB} = 10 \log_{10}(3) + 20 \log_{10} 2 - 20 \log_{10} \alpha$$

$$\text{SNR}_{dB} \approx 4.77 + 6.02R - 20 \log_{10} \alpha$$

$$3) \sigma = \frac{V}{\sqrt{2}} \Rightarrow \alpha = \frac{V}{\sigma} = \frac{V}{V/\sqrt{2}} = \sqrt{2}$$

$$\alpha = \sqrt{2} \Rightarrow \text{SNR}_{dB} = 4.77 + 6.02R - 20 \log_{10}(\sqrt{2})$$

$$\text{SNR}_{dB} = 1.76 + 6.02R$$

$$12) \text{Total levels } (L) = 6 \text{ segments} \times 16 \text{ levels/segment} = 96 \text{ levels}$$

$$\text{Number of bits } (n) \Rightarrow 2^n \geq L$$

$$\text{if } n=6, 2^6 = 64 (<)$$

$$\text{if } n=7, 2^7 = 128 (V) \quad n=7$$

$$5) \text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad [-a, a], P_s = \frac{(2a)^2}{12} \Rightarrow a=3$$

$$\Rightarrow P_s = \frac{(3 - (-3))^2}{12} = \frac{6^2}{12} = 3 \text{ W}$$

$$P_n \Rightarrow 16\Delta_1 + 16 \times 2\Delta_1 + 16 \times 4\Delta_1 = 30$$

$$16\Delta_1(1+2+4) = 3 \Rightarrow 112\Delta_1 = 3 \Rightarrow \Delta_1 = 0.0268$$

$$P_n = \frac{\Delta_1^2}{12} = 2 \times \sum_{i=1}^3 \int \frac{\Delta_1^2}{12} f_x(x) dx$$

$$= 2 \times \sum_{i=1}^3 \left(\frac{\Delta_1^2}{12} \cdot \frac{1}{6} \text{ W/Hz} \right) \text{ width } 16\Delta_1$$

$$P_n = \frac{2}{72} \times 16 \times (\Delta_1^3 + \Delta_2^3 + \Delta_3^3) = \frac{4}{9} (\Delta_1^3 + 2\Delta_1^3 + 4\Delta_1^3)$$

$$P_n = \frac{4}{9} \Delta_1^3 (1+2+4) = \frac{4}{9} \left(\frac{3}{112} \right)^3 \times 7$$

$$\approx 0.000626 \text{ W}$$

$$\text{SNR} = \frac{3}{0.000626} \approx 4792$$

$$\text{SNR}_{dB} \approx 36.8 \text{ dB}$$

$$1) i) V_{pp} = 1 - (-1) = 2V \quad L = 2^8 = 256$$

$$D = \frac{V_{pp}}{L} = \frac{2}{256} = \frac{1}{128} \approx 0.0078125V$$

$$\sigma_g^2 = \frac{\Delta^2}{12} = \left(\frac{1}{128} \right)^2 = \frac{1}{196608} \quad \sigma_g^2 = 5.086 \times 10^{-6} V^2$$

$$14) B = 15 \text{ KHz}$$

$$a) n = \log_2(L) = \log_2(65536) = 16 \quad n = 16 \text{ bits}$$

$$b) \text{ SQNR, } P = 0.1 \text{ W, } V_p = 1V$$

$$n = 16 \quad N_g = \frac{\Delta^2}{12} = \frac{\left(\frac{2V_p}{L} \right)^2}{12} = \frac{\left(\frac{2}{65536} \right)^2}{12}$$

$$\text{SQNR} = \frac{3 \cdot P \cdot 2^{2n}}{V_p^2} = 1288490.188$$

$$\text{SQNR}_{dB} = 91.09 \text{ dB}$$

$$15) W = 10 \text{ KHz} \quad L = 80 \text{ dB}$$

$$P_m = 16 \text{ W} \quad S_n(f) = \frac{N_0}{2} = 10^{-12} \text{ W/Hz}$$

$$|m(f)|_{\max} = 6$$

$$\text{SNR}_0 = 50 \text{ dB}$$

$$\Rightarrow N_0 = 2 \times 10^{-12} \text{ W/Hz}$$

$$a) \text{ DSB-SC-AM, } B_c = 2W = 2 \times 10 \text{ KHz} = 20 \text{ KHz}$$

$$P_c = \text{SNR}_0 = \frac{P_r}{N_0 W} = P_r = 10^5 \times 2 \times 10^{-12} \times 10^4$$

$$P_r = 2 \times 10^{-3} \text{ W}$$

$$P_c = P_r \times 10^8 = 2 \times 10^{-3} \times 10^8 = 2 \times 10^5 \text{ W} = 200 \text{ kW}$$

$$b) AM(m = 0.5)$$

$$SNR_0 = \frac{P_r M^2 P_{mn}}{N_0 W (1 + M^2 P_{mn})} = 6.00502$$

$$P_c = P_r \times 10^8 \approx 903 \text{ kW}$$

$$(6) P_s = \text{area} = \frac{1}{2} \pi 2W \times B = P_0 W$$

$$P_s = 2 \times P_0 W \text{ demodulation} = \frac{2 \times P_0}{4} = \frac{P_0}{2}$$

$$S_0 = \int_{-W}^W \frac{P_0}{2} \left(1 - \frac{|f|}{W}\right) df = 2 \times \frac{1}{2} W \times \frac{P_0}{2} = \frac{P_0 W}{2}$$

$$2) S_n \text{ baseband } (f) = \frac{1}{4} \left(S_{nn}(f-f_c) + S_{nn}(f+f_c) \right)$$

$$= \frac{1}{4} \left[\frac{N_0}{2} + \frac{N_0}{2} \right] = \frac{N_0}{4}$$

$$N_0 = \int_{-W}^W \frac{N_0}{4} df = \frac{N_0 W}{2}$$

$$(7) c(f) = 10 \cos(2\pi f_c t) \quad A_c = 10; \quad m = 0.5$$

$$\text{noise PSD, } \frac{N_0}{2} = 10^{-10} \text{ W/Hz} \Rightarrow N_0 = 2 \times 10^{-10} \text{ W/Hz}$$

$$m(t) = A_m \cos(2\pi f_m t)$$

$$m = \frac{A_m}{A_c} \Rightarrow A_m = m A_c = 0.5 \times 10 = 5$$

$$B_T = 2000 \text{ Hz}$$

$$\textcircled{1} P_s \Rightarrow s(t) = A_c (1 + M \cos(2\pi f_m t)) \cos(2\pi f_c t)$$

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c M}{2} \cos(2\pi(f_c + f_m)t) + \frac{A_c M}{2} \cos(2\pi(f_c - f_m)t)$$

total power

$$P_s = P_{\text{carrier}} + P_{\text{sideband}} = \frac{A_c^2}{2} + \frac{A_c^2 M^2}{4}$$

$$P_s = \frac{10^2}{2} + \frac{10^2 \times 0.5^2}{4} = 50 + 6.25 = 56.25 \text{ W}$$

$$B) P_n = P_n = 2 \times \frac{N_0}{2} \times B_T = N_0 B_T$$

$$P_n = 2 \times 10^{-10} \times 2000 = 4 \times 10^{-7} \text{ W}$$

$$SNR_0 = \frac{P_m}{P_{\text{noise}}} = \frac{A_c^2 M^2 / 2}{N_0 B} = \frac{A_m^2}{2 N_0 B} = \frac{5^2}{(2 \times 10^{-10}) (10^4)}$$

$$= 6.25 \times 10^7$$

$$SNR_0(\text{dB}) = 10 \log_{10}(6.25 \times 10^7) \approx 77.96 \text{ dB}$$